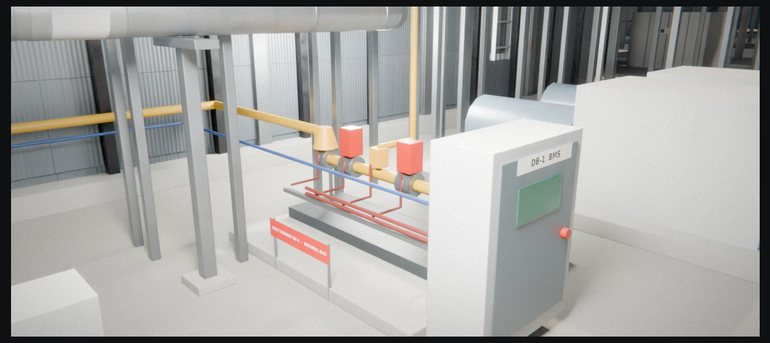
# BOILERS: GREENFIELD & BROWNFIELD

One cable partner from the new HRSG to the package boiler to the outage that replaces a module: the same products, staged and pulled the same way.



01 · GREENFIELD

## HEAT-RECOVERY BOILER

Behind every gas turbine: three-pressure HRSG with SCR and a duct burner. Feed pumps at 13.8 kV, drum-level and thermocouple runs, burner management in a Class I Div 2 skid.

**705**
CABLES PER HRSG TRAIN

**156,000**
FT OF CABLE PER TRAIN



02 · GREENFIELD OR RETROFIT

## GAS-FIRED PACKAGE BOILER

The plant's auxiliary boiler, the same package sold to industry: burner and BMS, gas train with double block and bleed, FD fan on a VFD, feedwater pumps, its own MCC e-house.

**83**
CABLES PER PACKAGE BOILER

**42,000**
FT OF CABLE



03 · BROWNFIELD

## MODULE REPLACEMENT OUTAGE

An HP evaporator module lifted out through the opened roof. The cable work rides the same outage window, so it has to be staged, pre-terminated and pulled fast.

**Outage**
CRITICAL PATH

**SIMpull**
+ PRE-TERMINATED HARNESSES

CABLE BY BOILER TYPE

| | MV-105 15 KV FEED PUMPS | TC-ER 600 V POWER & VFD | ARMOR-X MC-HL GAS TRAINS (C1D2) | CONTROL TC-ER | INSTRUMENTATION PLTC · TC EXT. | SIMPULL · REEL ROLLERS | PRE-TERMINATED HARNESSES |
|---|---|---|---|---|---|---|---|
| HRSG + DUCT BURNER | ● | ● | ● | ● | ● | ● | ● |
| PACKAGE BOILER | | ● | ● | ● | ● | ● | ● |
| REPLACEMENT OUTAGE | ○ | ● | ● | ● | ● | ● | ● |

● in scope ○ when the outage includes feed-pump or MV work

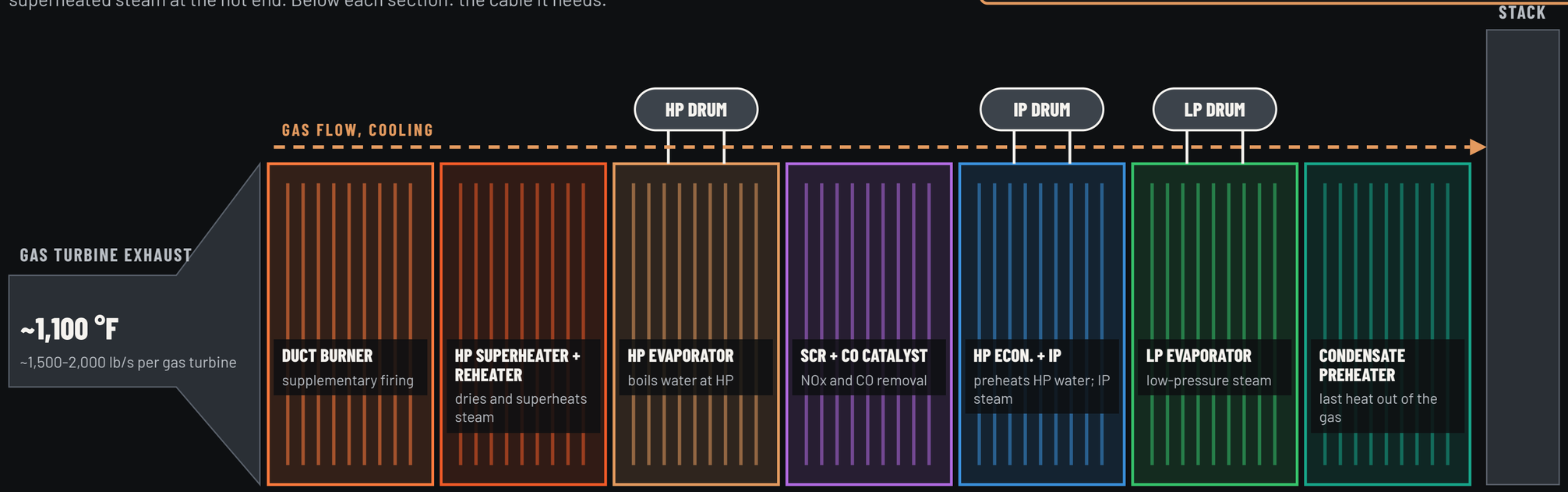
SW-3X1 MODEL · CONCEPTUAL ILLUSTRATION, NOT ENGINEERED · QUANTITIES FROM THE MODEL'S CABLE SCHEDULE

# HOW AN HRSG WORKS

The exhaust crosses rows of finned tubes, hottest sections first; water enters at the cold end and leaves as superheated steam at the hot end. Below each section: the cable it needs.

### WHY IT'S THERE

A gas turbine alone sends ~60 % of its fuel's energy up the stack as ~1,100 °F exhaust (~40 % efficient). The HRSG is a boiler heated by that exhaust: its steam drives the steam turbine, so the **same fuel makes about a third more power** (~60 %+ efficient; fuel, CO2 and cost per MWh down by about a third). It also houses the SCR that cuts NOx, and its duct burners can fire extra gas for peak output. **The bridge between the two halves of a combined cycle.**




Steam to the steam turbine, typical: HP ~1,800-2,400 psig, ~1,050 °F · reheat / IP ~400-600 psig, ~1,050 °F · LP ~50-80 psig

**~1/3**
of a combined-cycle plant's output comes from HRSG steam, with no extra fuel (duct firing aside)

**~60 %+**
combined-cycle efficiency, against ~40 % for the gas turbine alone

**3 x 3**
three pressures, three drums; each drum level measured three ways, typically voted 2-out-of-3

**~1 hour**
a hot start: about an hour: every start cycles the tubes, which is why tube-metal thermocouples matter

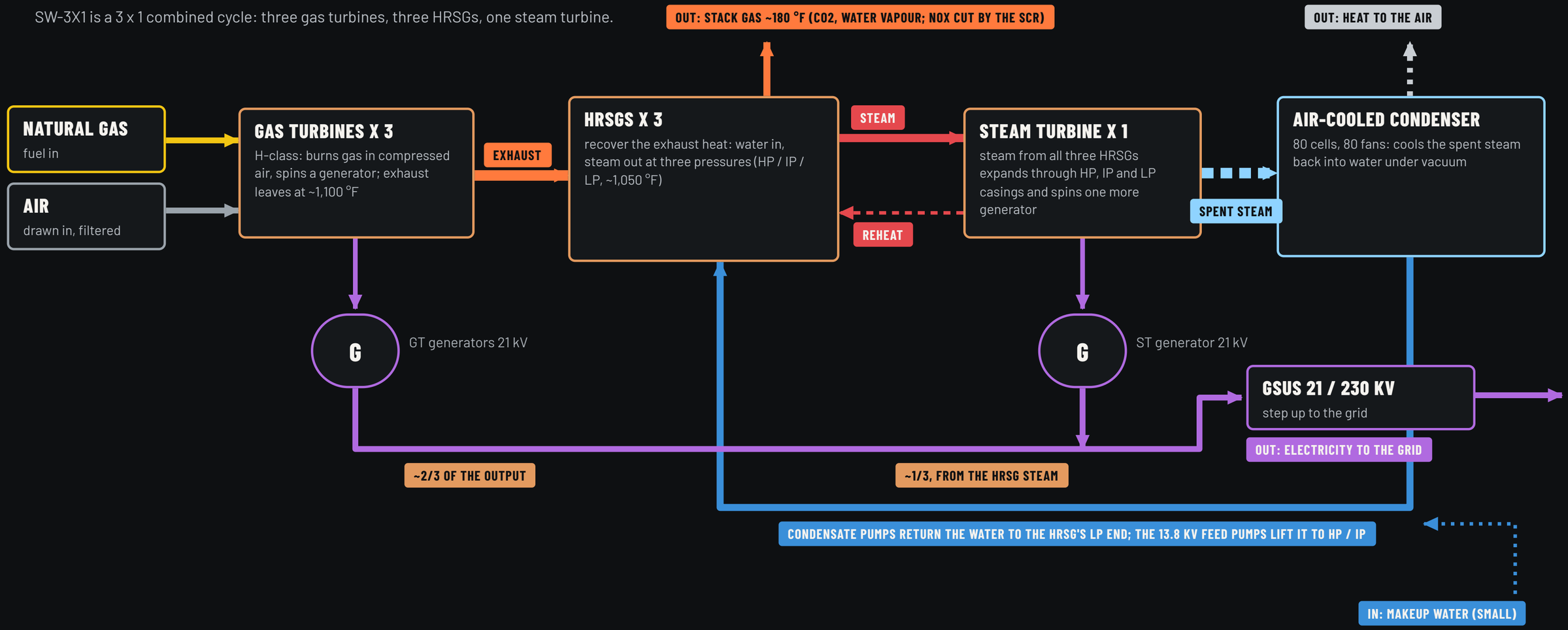
Typical three-pressure reheat HRSG with SCR behind an H-class gas turbine; section order, temperatures and pressures vary by design.



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# WHAT GOES IN, WHAT COMES OUT

SW-3X1 is a 3 x 1 combined cycle: three gas turbines, three HRSGs, one steam turbine.




### IS IT STILL STEAM?

Leaving the turbine it is spent steam: ~100-150 °F, below atmospheric pressure, part water already. The air-cooled condenser turns it back into water.

### A CLOSED LOOP

The same water goes round and round: pump, HRSG, turbine, condenser. Only a little makeup water replaces blowdown losses.

### WHERE THE POWER COMES FROM

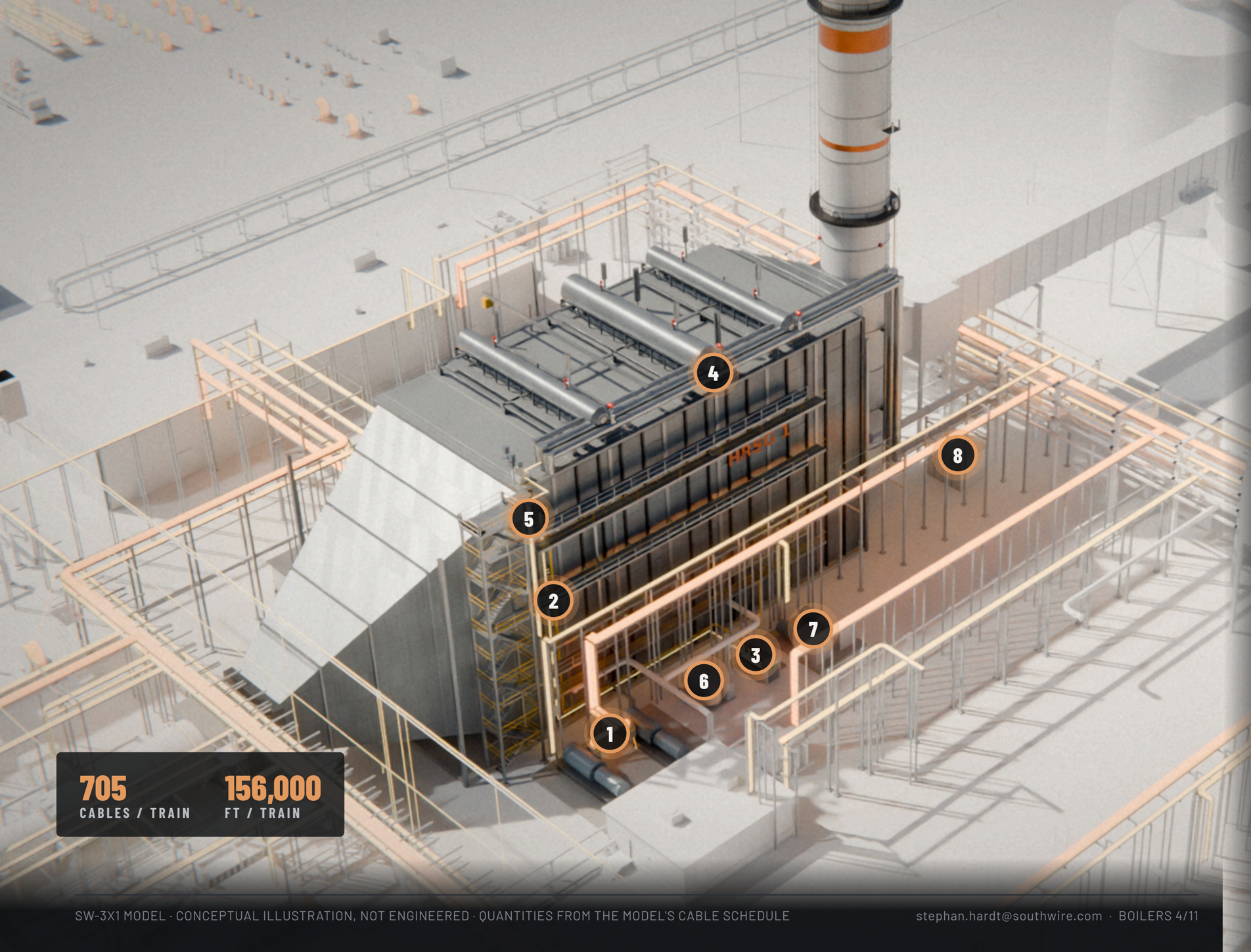
Fuel is burned only in the gas turbines (and duct burners). The steam turbine adds about half again on top, from heat that would otherwise go up the stack.

### THE CABLE ON THE LOOP

21 kV isolated-phase bus to the GSUs; MV-105 to the feed pumps; VFD cable to 80 condenser fans; instruments on every pump, valve and level.



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**705** CABLES / TRAIN

**156,000** FT / TRAIN

BOILER SOLUTIONS · HRSG

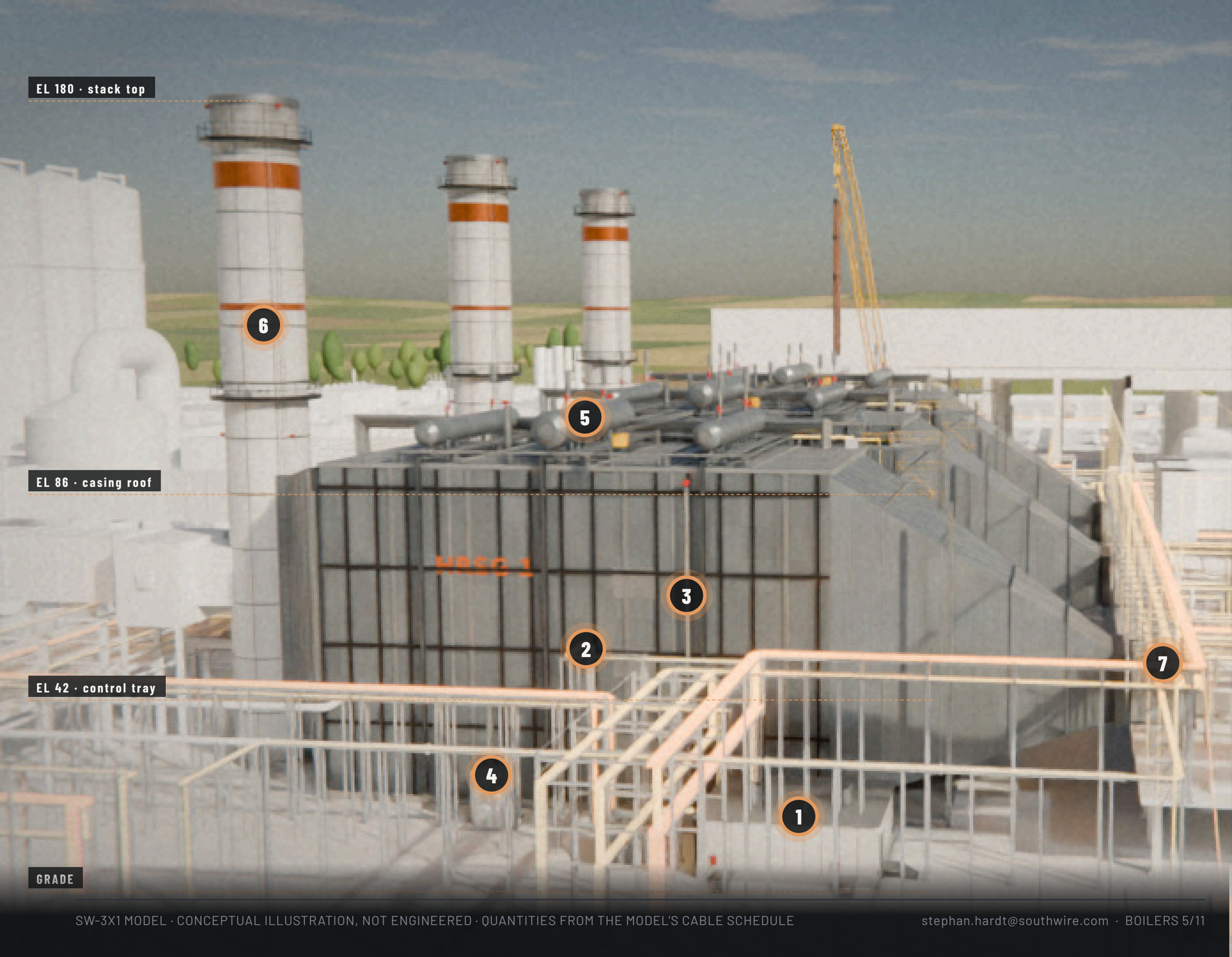
# THE HRSG, EVERY CABLE

One HRSG train and everything wired to it: cable in copper (MV) and amber (LV, control, instruments), the rest of the plant faded.

1. **FEED PUMPS, 13.8 KV**
   MV terminal boxes off the drop tray
   MV-105 15 kV [1]
2. **DUCT BURNER**
   gas header, 8 runners, igniters and flame scanners
   Control TC-ER [5] Instrumentation PLTC [6]
3. **DUCT-BURNER SKID AND BMS**
   double block and bleed, Class I Div 2
   ARMOR-X MC-HL [3] Control TC-ER [5]
4. **DRUMS AND ROOF VALVES**
   drum-level transmitters, motor-operated stop valves
   TC-ER power [2] Instrumentation PLTC [6]
5. **RISER TRAY TO THE ROOF**
   control, instrument and thermocouple cable
   Control TC-ER [5] Instrumentation PLTC [6] Thermocouple ext. [7]
6. **CHEMICAL FEED**
   metering-pump motors
   TC-ER power [2]
7. **SCR DILUTION-AIR BLOWERS**
   480 V motors
   TC-ER power [2] MC-HL VFD [4]
8. **CEMS SHELTER**
   analyzers and stack probes
   Instrumentation PLTC [6]



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BOILER SOLUTIONS · ELEVATION

# THE TRAIN IN ELEVATION

The same train at a low angle from the west: e-house, the tray into the marshalling box, the ammonia grid, drums and stack. Cable in copper and amber, the rest faded.

1. **HRSG E-HOUSE R2A**
   MCCs and DCS I/O for the unit
   TC-ER power [2] Control TC-ER [5]
2. **EL 42 CONTROL TRAY**
   into the west-wall marshalling box
   Control TC-ER [5] Instrumentation PLTC [6]
3. **AMMONIA INJECTION GRID**
   AIG header and lances, SCR
   Control TC-ER [5] Instrumentation PLTC [6]
4. **BLOWDOWN TANK**
   level and valve control
   Control TC-ER [5] Instrumentation PLTC [6]
5. **DRUMS AND ROOF VALVES**
   level transmitters, MOVs
   TC-ER power [2] Instrumentation PLTC [6]
6. **STACK AND CEMS PORTS**
   sample lines, probes
   Instrumentation PLTC [6]
7. **GT EXHAUST TO THE HRSG**
   inlet duct thermocouples
   Thermocouple ext. [7]

# INSIDE THE ROOM

Roof and two walls lifted off the HRSG 1 e-house: every cable leg, transformer to section.

1. **13.8 KV PRIMARY**
   MV-105 off the EL 36 tray into T-R2A · 3 cables, 660 ft
   **MV-105 15 kV** [1]
2. **480 V SECONDARY**
   10 sets of 4 x 1/C 500 kcmil · 40 cables, 4,800 ft
   **XHHW-2 1/C 500 kcmil**
3. **MAIN BREAKER**
   the 40 cables drop into LC-R2A
   **XHHW-2 1/C 500 kcmil**
4. **LC-R2A FEEDERS**
   to MCCs, VFDs, UPS and panels
   **TC-ER power** [2]
5. **MCC DROPS**
   power and control into each bucket
   **TC-ER power** [2] **Control TC-ER** [5]
6. **VFD AND MCC-2**
   SCR blowers, chemical feed, blowdown
   **TC-ER power** [2] **MC-HL VFD** [4]
7. **MARSHALLING**
   home-run multipairs from the field JBs
   **Instrumentation PLTC** [6] **Thermocouple ext.** [7]
8. **OUT TO THE FIELD**
   power via the north wall, instruments via the roof
   **TC-ER power** [2] **Control TC-ER** [5] **Instrumentation PLTC** [6]
9. **UPS, DC AND PANELS**
   125 V DC, 120 V AC UPS, lighting
   **TC-ER power** [2] **Control TC-ER** [5]




Inside: LC-R2A and MCC-R2A-1 (right), VFDs (left), trays overhead



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# SWITCHGEAR ROOM TO THE HRSG

Every leg, from the 13.8 kV breaker to the field devices, by cable application: Southwire product (numbers as on the offering slide), cable count and footage from the model's schedule.

**GSU-1 · UAT-1**
21 / 13.8 kV



SWITCHGEAR ROOM R1

**SWGR-13.8-1**

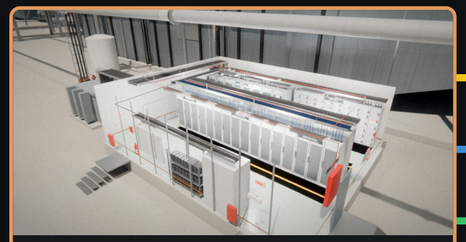
13.8 kV metal-clad: the feed-pump breakers and the T-R2A transformer feeder

MV-105 15 KV · 6 CABLES · 8,400 FT · 1

PRIMARY · MV-105 · 3 CABLES · 660 FT

T-R2A 13.8/ 0.48

SECONDARY · 1/C 500 KCMIL · 40 CABLES · 4,800 FT



HRSG E-HOUSE R2A

**LC-R2A · MCCS · VFDS · DCS**

480 V switchgear, starters, drives, UPS / DC, marshalling

In the room: lighting and receptacles, 2,280 ft THHN; panel feeder, control, fire-alarm loop, 330 ft

**MAIN CONTROL ROOM**
DCS servers, CEMS data

DCS FIBRE · 1 CABLE · 730 FT

**FEED PUMPS BFP-1A / 1B, 13.8 KV**
MV-105 15 kV · 6 cables · 8,400 ft 1

**MOTORS, MOVS, HEATERS: SCR BLOWERS, CHEMICAL FEED, BLOWDOWN, CEMS**
TC-ER 600 V power · VFD · 77 cables · 17,800 ft 2 · 4

**VALVES, MOVS, BURNER MANAGEMENT, LOCAL STATIONS**
Control TC-ER · ARMOR-X MC-HL · 96 cables · 21,140 ft 5 · 3

**FIELD JUNCTION BOXES TO TRANSMITTERS, FLAME SCANNERS, RTDS**
Instrumentation PLTC / ITC · RTD · 439 cables · 86,600 ft 6

**GAS-PATH, TUBE-METAL AND DUCT THERMOCOUPLES**
Thermocouple extension · 84 cables · 19,680 ft 7

**T-R2A FANS, CONTROLS, TEMPERATURE / GAS MONITORING**
TC-ER power · control · PLTC · 38 cables · 3,600 ft 2 · 5 · 6

**CEMS SHELTER: DATA TO THE CONTROL ROOM, DETECTION, LIGHTING**
Fibre · fire alarm · THHN · 3 cables · 2,580 ft —

THE HRSG 1 TRAIN: 705 CABLES, ~156,000 FT

+ FEEDING THE ROOM: 86 CABLES, 12,400 FT




R2A also serves the GT1 unit loads, so the T-R2A primary and secondary, the transformer auxiliaries, the DCS fibre and the room's own wiring are shown here but are not in the train total. Schedule rows: A-MV-0015, A-P-0422 and the rest by tag.

SW-3X1 MODEL · CONCEPTUAL ILLUSTRATION, NOT ENGINEERED · QUANTITIES FROM THE MODEL'S CABLE SCHEDULE

# LEVELS, RATINGS AND THE CABLE THEY DRIVE

One HRSG train from the generator down to the field signals: each voltage level, the equipment and its rating, what it feeds, and the cable that follows.

| LEVEL | EQUIPMENT · RATING | FEEDS · CURRENT | CABLE IT DRIVES |
|---|---|---|---|
| 230 kV | **GSU-1, 21 / 230 kV step-up** | Plant to the grid, overhead line | **Overhead conductor (outside the HRSG scope)** |
| 21 kV | **GT1 generator, isolated-phase bus; UAT-1 21 / 13.8 kV, 40 MVA** | UAT-1 secondary 1,674 A into SWGR-13.8-1 | **4 sets of 3 x 1/C 750 kcmil MV-105** |
| 13.8 kV | **SWGR-13.8-1: metal-clad, 1,200 A, 40 kA, arc-resistant** | Feed pumps BFP-1A / 1B, 5,000 kW each, direct-on-line, 232 A; T-R2A primary, 105 A | **3 x 1/C 250 kcmil MV-105 per pump; 3 x 1/C #2 to T-R2A** |
| 480 V | **T-R2A 2,500 kVA → LC-R2A 4,000 A main, 65 kA → MCC-R2A-1 / -2, 800 A, 65 kA; VFDs** | Secondary 3,007 A; SCR blowers 2 x 110 kW (156 A); 45 HRSG MOVs; feed-pump and chemical-feed auxiliaries; welding receptacles | **10 sets of 4 x 1/C 500 kcmil; 3/0 TC-ER / MC-HL VFD; #10 TC-ER to each MOV** |
| 120 V AC / 125 V DC | **UPS-R2A 10 kVA; 125 V DC battery, 60 cells; DC and UPS panels** | DCS, F&G node, breaker trip and close, burner-management power | **1/C #6 and control TC-ER 14 AWG** |
| Signals | **4-20 mA / HART, thermocouple millivolts, 24 V DC discretes** | Drum level, pressures, flame scanners, gas-path and tube-metal temperatures | **439 instrument + 84 thermocouple-extension cables, shielded pairs** |

**750 kW**
connected at 480 V on one HRSG train, 77 circuits

**2 x 5 MW**
feed pumps: the largest motors on the train, at 13.8 kV

**3,007 A**
on the T-R2A secondary: 40 cables of 500 kcmil

**74 %**
of the train's cables are instrument or thermocouple pairs (68 % of the footage)

**8.4 / 24.6**
cal/cm$^2$ arc flash at 13.8 kV / 480 V (equipment labels)

Ratings are the model's typical values (Load List and nameplates), not engineered. Real plants vary: many put the feed pumps on a 4.16 or 6.9 kV bus.



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# WHAT THE SPEC CALLS FOR

Every cable application on the HRSG, the requirements an owner's or EPC's cable specification writes for it, and the Southwire product built to meet them.

| APPLICATION · WHERE | WHAT THE SPECIFICATION CALLS FOR | SOUTHWIRE |
|---|---|---|
| **MV POWER, 15 KV**<br>FEED-PUMP MOTORS; THE T-R2A PRIMARY | 1/C Cu, EPR insulation, **133 % level**, 90 °C, tape shield, PVC jacket. ICEA S-93-639, UL 1072 (MV-105). Terminations IEEE 48; acceptance test VLF (IEEE 400.2) / NETA ATS | **MV-105 15 kV, 220 mil NL-EPR 133 %, SIMpull · SPEC 46302** |
| **480 V FEEDERS AND POWER**<br>T-R2A SECONDARY, LC-R2A TO MCCS, MOTORS AND MOVS | Cu XHHW-2, 90 °C wet / dry, sized at **125 %** of full load; voltage drop ≤ 3 %. ICEA S-95-658, UL 44 / UL 1277; 10+ parallel sets: compare with bus duct | **XHHW-2 1/C (SIMpull) · Type TC-ER 3/C + ground · SPEC 45252** |
| **TRAY CABLE: POWER AND CONTROL**<br>EVERY TRAY BETWEEN THE E-HOUSE AND THE HRSG | Type TC-ER (UL 1277) so it may leave the tray as an exposed run to the equipment (NEC 336.10); sunlight resistant, direct burial; **IEEE 1202 / UL 1685** 70,000 Btu/h vertical-tray flame test | **Control TC-ER 14 AWG multiconductor · SPEC 45053** |
| **HAZARDOUS AREA: GAS TRAIN, DUCT BURNER**<br>DUCT-BURNER FUEL SKID, BMS PANEL TO THE VALVES | Class I Div 2 wiring methods per NEC 501.10(B); MC-HL (UL 1569 + UL 2225) also qualifies for Div 1. Burner management per **NFPA 85**, duct burner UL 295, listed fittings | **ARMOR-X MC-HL, welded armor · SPEC 45352** |
| **VARIABLE-FREQUENCY DRIVES**<br>SCR DILUTION-AIR BLOWERS, FD FANS | VFD-rated: symmetrical three-ground construction, overall shield or continuous armor for the common-mode current, XLPE insulation for reflected-wave peaks | **ARMOR-X MC-HL VFD · TC-ER VFD · SPEC 45352** |
| **INSTRUMENTATION, 4-20 MA / HART**<br>DRUM LEVEL, PRESSURES, FLAME SCANNERS, ANALYZERS | **Individually shielded** twisted pairs + overall shield, drain wires; 300 V PLTC (UL 13, NEC 725) or ITC; ICEA S-73-532; segregated from power in tray | **Instrumentation PLTC, SPOS · SPEC 43303** |
| **THERMOCOUPLE EXTENSION**<br>GAS-PATH, TUBE-METAL AND DUCT TEMPERATURES | Alloy matched to the thermocouple (type K = KX), ANSI MC96.1 / ASTM E230 colour code; **never copper**; shielded pairs; PLTC; 105 °C near the casing | **Thermocouple extension, PLTC multi-pair · SPEC 42345 / 42350** |
| **FIRE ALARM, FIBRE, BUILDING WIRE**<br>F&G LOOPS, CEMS DATA, E-HOUSE LIGHTING | FPLR / FPLP (NEC 760, NFPA 72); single-mode fibre, armored in tray; THHN / THWN-2 in conduit (UL 83) | **SIMpull THHN / THWN-2 · fire-alarm and fibre cable** |

## CHECKED AGAINST A REAL PLANT

A recent US utility combined-cycle filing shows the same structure: **HRSG MCCs A / B** fed from 480 V switchgear through 2,500 / 3,300 kVA transformers off the auxiliary bus (6.9 kV there, 13.8 kV here).

## DESIGN AND INSTALL

Raceway and cable design for generating stations: IEEE 422. Grounding: IEEE 80 / 665. Area classification: NEC 500, API RP 500. Field acceptance: NETA ATS.



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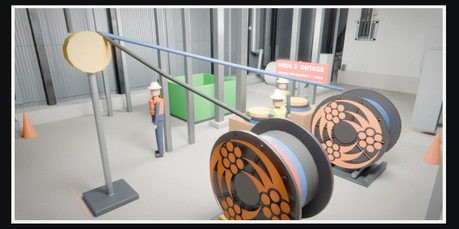
HRSG 3: an HP evaporator module lifted out through the opened roof by a crawler crane; the new module waits on a lowboy on the ring road.

BOILER SOLUTIONS · REPLACEMENT

# THE OUTAGE CLOCK IS THE COST

On a boiler or HRSG replacement every hour of the outage costs. Cable rides the same window, so it moves off the critical path.

- **STAGE BEFORE THE TRIP** cut-to-length reels on SIMpull Cradles and reel rollers, labelled per pull, at the unit before the outage starts
- **PRE-TERMINATE** factory-made harnesses with QR tags land on the marshalling boxes: test, not terminate, on the critical path
- **PULL CLEAN** SIMpull reels and a sheave at the HRSG wall: one crew, no reel jacks, tension recorded
- **CUT BACK AND RECYCLE** old cable segregated into recycling bins as the module comes out
- **CONVERSIONS** new burner management and gas trains in Class I Div 2 run in ARMOR-X MC-HL



Cable work at the HRSG e-house

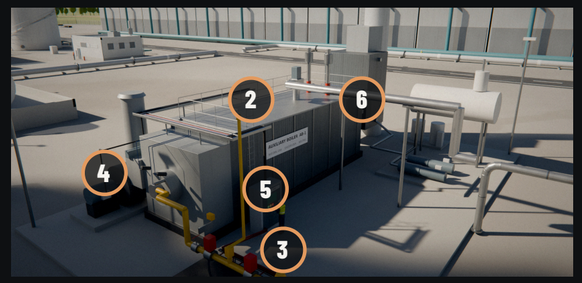


Old module on the hook

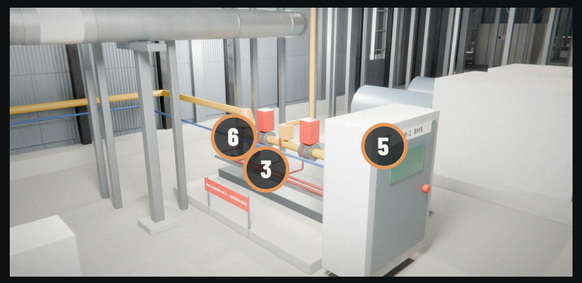


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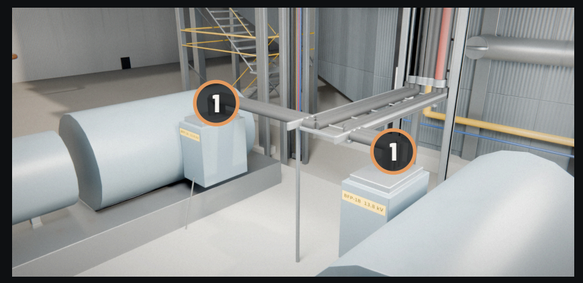
# THE CABLE AT EVERY SPOT



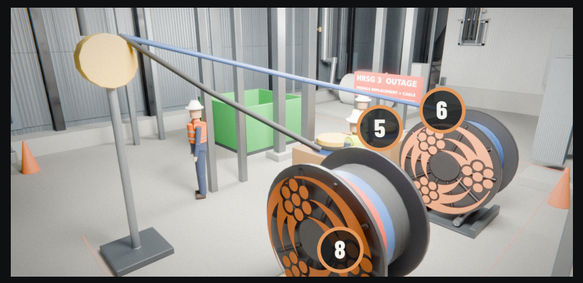

Package boiler AB-1




HRSG duct-burner skid




HRSG feed pumps, 13.8 kV




Outage cable work, HRSG 3

| | PRODUCT | WHERE IT RUNS | SPEC | SOUTHWIRE STOCK # · SIZE | SPEC SHEET |
|---|---|---|---|---|---|
| 1 | **MV-105 15 KV**<br>1/C Cu, 220 mil NL-EPR 133 %, tape shield, SIMpull PVC | Feed-pump motors, 13.8 kV | 46302 | 95604599 · 500 kcmil | southwire.com › |
| 2 | **TYPE TC-ER POWER**<br>3/C Cu XHHW-2 + ground, PVC, 600 V | Fans, pumps, MCC feeders | 45252 | 95305999 · 12 AWG 3/C | southwire.com › |
| 3 | **ARMOR-X MC-HL**<br>Cu XLPE XHHW-2, welded armor, PVC, 600 V, Class I Div 2 | Gas trains, burner skids | 45352 | 89052799 · 8 AWG 4/C | southwire.com › |
| 4 | **ARMOR-X MC-HL VFD**<br>3/C Cu XLPE XHHW-2 + ground, VFD rated | FD fan on its drive | 45352 | 89051899 · 3/0 3/C | southwire.com › |
| 5 | **CONTROL TC-ER**<br>Cu XLPE XHHW-2, 600 V | BMS panels, valves, MOVs | 45053 | 89067701 · 14 AWG 5/C | southwire.com › |
| 6 | **INSTRUMENTATION PLTC / ITC**<br>300 V, shielded pairs + overall shield (SPOS) | Flame scanners, drum level, transmitters | 43303 | per pair count | southwire.com › |
| 7 | **THERMOCOUPLE EXTENSION**<br>PVC, PLTC multi-pair (KX on request), 105 °C | HRSG gas-path and tube temperatures | 42345 | per construction | cabletechsupport.southwire.com › |
| 8 | **SIMPULL CRADLE**<br>for 51 / 61 in SIMpull Reels, one-person payoff | Outage staging and pulls | SPR-CRDL | 67606001 | southwire.com › |

Stock numbers are one catalogue size per product, from Southwire's product listings; every size and construction is on the same SPEC. Confirm current stock with Southwire customer service before quoting.



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